$(2x - y + 1)dx - (x + 3y - 2)dy = 0$

$\frac{\partial M}{\partial y} = -1 \qquad \frac{\partial N}{\partial x} = -1$

$\frac{\partial u}{\partial x} = (2x - y + 1)dx$

$u = x^2 - xy + x + g(y)$

$\frac{\partial u}{\partial y} = -x\,dy + g'$

$(\cancel{-x} - 3y + 2)dy = \cancel{-x} + g'$

$(-3y + 2)dy = +g'$

$g = -\frac{3}{2}y^2 + 2y$

$u = x^2 - xy + x - \frac{3}{2}y^2 + 2y = c$

$2x^2 - 2xy + 2x - 3y^2 + 2y = k$

1) $(2x + y^2)dx + 2xydy = 0$

$\frac{\partial M}{\partial y} = 2y \qquad \frac{\partial N}{\partial x} = 2y$

$\frac{\partial u}{\partial x} = (2x + y^2)dx$

$u = x^2 + xy^2 + g(y)$

$\frac{\partial u}{\partial y} = 2xy\,dy + g'$

$\cancel{2xy}dy = \cancel{2xy}dy + g'$

$g = 0$

$x^2 + xy^2 = c$

2) $(2x-1)dx + (3y+7)dy = 0$

$$\frac{\partial M}{\partial y} = 0 \qquad \frac{\partial N}{\partial x} = 0$$

$$\frac{\partial u}{\partial x} = (2x-1)\,dx$$

$$u = x^2 - x + g(y)$$

3) $5x^3 + 8xy - 4y^4 = k$

$$\frac{\partial u}{\partial y} = g'$$

$$3y + 7 = g'$$

$$g = \frac{3y^2}{2} + 7y$$

$$x^2 - x + \frac{3y^2}{2} + 7y = k$$

$$2x^2 - 2x + 3y^2 + 14y = c$$

3) $(5x+4y)dx + (4x-8y^3)dy = 0$

$$\frac{\partial M}{\partial y} = 4 \qquad \frac{\partial N}{\partial x} = 4$$

$$\frac{\partial u}{\partial x} = (5x+4y)\,dx$$

$$u = \frac{5x^2}{2} + 4xy + g(y)$$

$$\frac{\partial u}{\partial y} = 4x\,dy + g'$$

$$(\cancel{4x} - 8y^3)\,dy = 4x\,\cancel{dy} + g'$$

$$g' = -8y^3\,dy$$

$$g = -2y^4$$

$$\frac{5x^2}{2} + 4xy - 2y^4 = k$$

$$5x^2 + 8xy - 4y^4 = C$$

10) $\left[y\cos(xy) + \frac{y}{\sqrt{x}}\right]dx + \left[x\cos(xy) + 2\sqrt{x} + \frac{1}{y}\right]dy = 0$

$$\frac{\partial M}{\partial y} = \cos(xy) - xy\sin(xy) + \frac{1}{\sqrt{x}}$$

$$\frac{\partial N}{\partial x} = \cos(xy) - xy\sin(xy) + \frac{1}{\sqrt{x}}$$

$$\frac{\partial u}{\partial x} = \left(y\cos(xy) + \frac{y}{\sqrt{x}}\right)dx$$

$$u = \sin xy + 2y\sqrt{x} + g(y)$$

$$\frac{\partial u}{\partial y} = x\cos xy\,dy + 2\overset{dy}{\sqrt{x}} + g'$$

$$\left(x\cos\cancel{(xy)} + 2\cancel{\sqrt{x}} + \frac{1}{y}\right)dy = x\cos\cancel{x}y\,dy + 2\cancel{\sqrt{x}}\,dy + g'$$

$$g' = \frac{dy}{y}$$

$$g = \ln y$$

$$\sin xy + 2y\sqrt{x} + \ln y = C$$